Joshita Arora

+1-(352) 709-0788 | joshitaarora7@gmail.com | linkedin.com/in/joshita-arora

Work Experience

Machine Learning Researcher, *Gainesville, FL, USA* May 2025 – May 2026

- Owned end-to-end ML pipeline for astronomical object classification from Euclid telescope imagery, including data preprocessing, experiment tracking, model benchmarking, validation strategy, and deployment support across large image datasets.
- Evaluated ConvNeXT, DenseNet, ResNet, DeiT, and EfficientNet architectures using controlled validation workflows, targeted augmentation, and failure analysis, improving classification accuracy to 96% and supporting downstream cosmological analysis.

AI Engineer, UF IFAS Extension, *Gainesville, FL, USA* Jun 2025 – May 2026

- Designed and deployed production-grade AI agent systems in Microsoft Copilot to automate administrative decision workflows using prompt orchestration, context engineering, evaluation loops, and compliance guardrails, improving production reliability, reducing failure modes, and enabling auditable decision workflows in high-stakes operational systems
- Built monitoring and feedback mechanisms for agent performance and workflow efficiency; key systems include funding classification (gifts vs. grants) with sub-5-minute turnaround and a position description generator that reduced hiring cycle time by 50%.

Senior Data Scientist, Viscadia, *Gurugram, India* Dec 2023 – Jul 2024

- Executed the firm's inaugural market research project, establishing conventions for future initiatives, while leading the accelerated onboarding process for new team members; trained 40+ team members on MR fundamentals.
- Delivered conjoint analysis for breakthrough oncology drug using SAS and R with 100+ patient simulations, completing project in 60% of expected timeframe and providing market share predictions that supported successful US drug approval.

Data Scientist (Market Research), ZS Associates, *Gurugram, India* Sep 2021 – Jul 2023

- Executed 10+ global pharma market research projects including chart audit, conjoint analysis, and demand estimation; analyzed large-scale healthcare datasets using SQL, R, and statistical methods to identify market trends and inform product strategy.
- Presented insights to cross-functional stakeholders, influencing pricing and launch decisions for multi-million-dollar products

Education

University of Florida, USA ***Outstanding Masters Award Recipient***
Master of Science, Computer Science (Thesis Track) GPA: **3.95/4.00**
Machine Learning, Distributed Systems, Advanced Data Structures, Database Systems, NLP, Statistical Learning, DBMS, Software Engineering

Indira Gandhi National Open University, India GPA: **9.76/10.00**
Post Graduate Diploma, Applied Statistics

Punjab Engineering College, India ***Administrator's Gold Medalist***
Bachelor of Technology, Mechanical Engineering GPA: **9.07/10.00**
Minor Specialization in Computer Science GPA: **9.40/10.00**

Technical Skills

Languages: Python, SQL, R, Java, JavaScript, Julia, MATLAB, SAS
Data Science: A/B Testing, Statistical Modeling, Experiment Design, Causal Inference, Hypothesis Testing, Data Analysis
ML/Systems: ML Pipelines, Model Monitoring, Experiment Tracking, Distributed Systems, Production Debugging
Tools & Platforms: Git, Docker, AWS, Linux, Pandas, NumPy, sklearn, TensorFlow, PyTorch, Copilot, Claude, Codex, Cursor

Projects and Research

- **Master's Thesis:** Reformulated transformer attention as an instance-specific linear operator to enable closed-form analysis of robustness, uncertainty propagation, and failure behavior under latent perturbations in neural networks.
- **BrainSafe:** Engineered an AI-powered browser extension using Google ADK and parallel agent workflows to enhance video consumption with real-time contrast/speed adjustments and content moderation for children.
- **Self-Organizing Maps-Topologies for Dark Energy Survey Data:** Implemented and benchmarked SOM with flexible topologies, exploring clustering performance improvements on DES Y6 data; released open-source python package on PyPI.
- **The Thinker:** Developed Abstract Syntax Tree-based knowledge base for semantic code search, preserving structural and contextual relationships to enhance retrieval accuracy using NLP techniques; improved performance over traditional methods.

Awards and Honors

- **Outstanding Masters and Professional Award (2026)**: UF International Center; recognized for academic excellence, advancing global citizenship, and contributing to UF's research, teaching and service missions.
- **Glassy Gator Award (2025)**: Dean of Extension; Artificial and Business Intelligence team recognized for impactful AI-driven service innovation across the university.
- **ShellHacks (2025), Google Cloud Challenge**: BrainSafe; winner among 175+ teams at Florida's largest hackathon.